

Smart Care Solution for Smart Home Devices

Sana Fatima Sayyed*

Computer Science

University of Nottingham

Nottingham, United Kingdom

psxss35@nottingham.ac.uk

Neelima Sailaja

Horizon Digital Economy Hub

University of Nottingham

Nottingham, United Kingdom

neelima.sailaja@nottingham.ac.uk

Abstract

The limited repairability and short lifecycle of smart home devices is contributing to the growing global electronic waste (e-waste) issue. Despite their convenience, these smart devices are frequently discarded due to high repair costs, complex proprietary designs, and restricted access to necessary repair resources. This study presents the design of a mobile application aimed to enhance repairability and lifecycle management for smart home devices. Through a user-centered design approach, which included surveys and participatory design sessions with users, this research identifies common barriers to device repair and integrates user insights to guide iterative app development. Key features such as a centralised dashboard, diagnostic tools, repair tutorials, and community support, address user needs while promoting sustainable everyday technology practices. By empowering users to manage and extend the lifespan of their devices, this application is expected to contribute to reducing e-waste by aligning with circular economy principles for sustainable technology use.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI).

Keywords

Smart home devices, user-centered design, circular economy, sustainability, e-waste, user centered design, AI enabled predictive analysis, voice-assistant

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1 Introduction

Electronic waste (e-waste) is the fastest-growing waste stream worldwide, with UK ranking as the second-highest generator per capita [16]. Global e-waste generation is projected to reach 82 billion kilograms by 2030 [1]. Here, smart home devices, such

*Corresponding author

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as thermostats, lighting systems, and security cameras become a key contributor. They provide convenience but their limited repairability due to high repair costs, complex proprietary designs, and limited access to spare parts lead users to replace rather than repair these devices, fostering a culture of disposability, accelerating e-waste generation [1] and leading to serious environmental and health consequences due to improper disposal of e-waste [3],[9].

Research shows that prolonging a product's use reduces its environmental impact and conserves resources [27]. Extending product lifecycles can significantly slow down consumption and mitigate e-waste generation [27]. Recognising the environmental impact, initiatives such as the "Right to Repair" (R2R) advocate for users' access to repair tools and spare parts to extend product lifecycles [26]. However, the R2R movement currently covers only certain household appliances, leaving smart home devices with limited repair options [21]. Many of these devices are designed with proprietary components or software, further restricting repairability [32]. There remains a critical gap in systems facilitating repairability and lifecycle management for smart home devices.

This study addresses this gap by designing a mobile application aimed at enhancing repairability and lifecycle management for smart home devices. Using an iterative design process driven by user feedback, the app provides tools, information, and connectivity to facilitate device repair, fostering user engagement in sustainable consumption practices and contributing to e-waste reduction. The application integrates key features such as step-by-step troubleshooting guides, access to spare part information, and community-driven repair assistance to support users in maintaining and repairing their devices. By empowering users with accessible repair resources and fostering a shift towards product longevity, the solution contributes to broader sustainability and circular economy goals. While it cannot eliminate systemic barriers, it serves as a bridge between users, technicians, manufacturers, and repair communities to create a more repair-friendly ecosystem.

2 Global E-Waste problem

The consumption of electrical and electronic equipment (EEE) has surged in the 21st century, driven by economic growth, industrialisation, and technological advancements. Most households now own multiple devices like refrigerators, washing machines, and smartphones, and nearly half of the global population is connected to the internet [19]. As a result, e-waste, a discarded electrical and electronic equipment, has become the fastest-growing waste stream worldwide. According to the Global E-Waste Monitor 2024, global e-waste production has risen from 34 billion kilograms in 2010 to 62 billion in 2020, with projections of 82 billion by 2030 [1]. E-waste contains valuable materials like copper, aluminium, and gold, but their improper disposal releases toxic substances such as

lead and mercury into the environment, posing serious health risks, particularly to vulnerable populations like children and pregnant women [5],[6],[7],[12].

As the world undergoes digital transformation, the number of connected devices, ranging from household appliances to smart gadgets, has been rapidly increasing [19]. The Internet of Things (IoT) is the significant part of this transformation. IoT refers to a system of interconnected physical objects, such as vehicles, appliances, and other items, that are equipped with sensors, software, and network capabilities, enabling them to gather and exchange data [32]. This sector is growing rapidly with global device projections rising from 15.9 billion in 2023 to more than 32.1 billion by 2030 [28]. Smart devices which are a subset of IoT and the focus of this study, include devices such as smartphones, smart TVs, smart washing machines, smart refrigerators, and smart sensors [32]. These everyday devices are enhanced with advanced computing technologies, allowing them to connect to networks and contribute to the IoT ecosystem. They integrate hardware and software components, such as home appliances, sensors, controllers, and communication networks that allows users to control household appliances remotely, and manage energy use more efficiently, thereby increasing convenience and comfort while reducing the effort required for performing daily tasks [14],[15]. In 2023, there were over 3.8 billion connected smart home devices worldwide, with approximately 19 million in use in the UK [29]. In the United Kingdom (UK), the household penetration rate of smart home automation stands at 11.4% and is expected to reach 44.9% by 2028 [15]. The rise of these devices contributes significantly to the e-waste problem [16], largely due to their lack of repairability and planned obsolescence [4], the intentional limitation of device longevity through design, resulting in a high replacement rate and significant e-waste [27]. This increasing adoption of smart technologies underscores the urgent need for sustainable solutions to manage and extend device lifecycles.

3 Circular Economy and repair

The circular economy (CE) offers a sustainable alternative to the traditional “take-make-dispose” model by keeping products and materials in use and thereby reducing waste. For managing electronic waste (e-waste), CE provides effective strategies to extend product lifecycles, minimise pollution, and conserve resources [3],[17]. This approach encourages essential shifts in production and consumption, aiming to eliminate waste and restore natural resources. Key CE practices include maintenance, reuse, refurbishment, remanufacturing, and recycling, which keep materials in circulation for longer [5], [30].

An essential element of the circular economy is product repair, which directly restores a product’s functionality, extending its lifespan without requiring the need for new materials [30]. Research indicates that repair can reduce environmental impact more effectively than recycling or remanufacturing by enabling direct reuse [3], [7]. By prolonging the service life of the device, repair supports CE principles, delaying products’ entry into the waste stream [3]. Additionally, extending the lifespan of smart products, to mitigate the environmental impact of IoT, is a key strategy that aligns with the United Nations Sustainable Development Goals [3]. Initiatives such as the UK’s Right-to-Repair (R2R) law and the EU’s Circular

Economy Action Plan aim to improve repairability by mandating manufacturers to make spare parts available for up to 10 years and to design repair-friendly products. However, these policies currently focus on major household appliances, often excluding IoT devices like smart home technologies [21].

Companies like Fairphone, known for its modular smartphones, and community-driven platforms like iFixit and Repair Cafes, play vital roles in fostering a repair culture. Their initiatives highlight the importance of accessible repair services, user-friendly designs, and readily available spare parts to support CE objectives. Research into repair processes at repair cafés provides valuable insights into the challenges repairers faced and offers insights for designing products that support repair [3], [30]. However, these studies often overlook smart devices, which pose unique difficulties due to complex software, proprietary components, and limited access to diagnostic tools. Research also suggests that repair strategies such as fault indication systems, user workshops, repair discounts, and trade-in programs, can further extend product lifespans, reduce e-waste, and lower carbon emissions. Engaging both consumers and stakeholders in repair practices is essential for advancing circular economy goals and achieving sustainable device management [13],[8],[18].

3.1 Barriers to Repair

Despite their benefits, users often opt for replacement over repair due to several challenges. These include diminishing satisfaction over time, frequent product upgrades [31], and a lack of modularity, all of which foster a culture of disposability and contribute to e-waste generation [12],[20],[25],[32]. High repair costs, limited availability of affordable spare parts, and insufficient manufacturer support for older models further discourage repairs [14],[18],[23],[25]. Additionally, restricted access to repair information, parts, and tools complicates the repair process [12],[16],[25] while the lack of accessible diagnostic tools, repair services, and DIY (Do It Yourself) hubs further limits repair options [25]. Studies show widespread dissatisfaction with professional repair services, with concerns over quality and repair times [25]. In the U.S., 40% of consumers worry about repair quality, while in the EU, 28% express dissatisfaction with repair times [12]. However, research suggests strong consumer interest in repairability with 77% of consumers willing to repair their devices if affordable services were available, and a similar percentage of Europeans are open to repair when the process is simple and cost-effective [27].

3.2 HCI and Repair

This paper focuses on implementing HCI principles to develop a systemic platform that encourages users to adopt sustainable practices, specifically focusing on repair as a means of extending product life. As shown in [13], Sustainable Human-Computer Interaction (HCI) has become vital in addressing technology’s ecological footprint by employing two main approaches: sustainability through design and sustainability in design. The former incorporates sustainable practices directly into the design process by using eco-friendly materials, enhancing energy efficiency, and extending product lifecycles [10], [13]. The latter approach leverages interactive systems to promote

environmentally responsible behaviours, such as conserving energy and mindful consumption, among users [10], [13].

In addition, past research in HCI has addressed issues of obsolescence, examining how user interactions can influence pathways toward extending product lifespans [22]. However, the absence of standardised sustainable design guidelines presents a key challenge, especially for interconnected smart technologies. As smart devices become increasingly complex, managing data, hardware, and software coordination requires adaptable design guidelines that align with sustainability goals. Addressing these challenges necessitates collaborative efforts to develop robust regulatory frameworks, foster manufacturer support, conduct specialised research, and empower users to effectively tackle the repair challenges of smart devices [10].

The Circular Economy framework offers a solution by promoting the extension of product lifespans through repair, upgrade, and reuse [9], [11]. Recent studies, including the work done by Rudolf et al. (2022) in Germany, shed light on the challenges and provide recommendations for improving the repairability of electrical and electronic equipment (EEE) [24]. Tools such as the Repair Score System, EN 45554, and the French Repair Index have emerged to evaluate repairability across product categories [2]. Additionally, Right-to-Repair (RTR) initiatives advocate for designing repair-friendly products and equipping consumers with the resources necessary for self-repair or professional repair services, such as instructions, diagnostic tools, and spare parts [8]. However, there remains a significant gap in implementing a comprehensive systemic solution to support and encourage self-repair among users. Despite efforts by companies like Fairphone to encourage repairable products, the rate of repair activities remains low, with limited systematic support from manufacturers [27]. DIY repair is often dependent on individual motivation. Recent research in design and business emphasises repairability from a user perspective but lacks a comprehensive solution. Developing a cohesive system that connects users with repair resources and supports sustainable repair practices is essential to create a more circular economy and mitigate environmental impact [2]. The rest of this paper illustrates the genesis of design of such a system that bridges the gap between manufacturers, repair communities, technicians and end users. Central to this system is an intuitive mobile application that empowers users with step-by-step repair guides, access to repair tools and spare parts vendors, professional repair services, and Artificial Intelligence (AI)-driven diagnostic and maintenance support. By integrating these elements, the proposed system aims to foster sustainable repair practices, reduce e-waste, and contribute to the development of a more circular economy.

4 Research Design

User-centered design methods that leveraged a suite of Human Computer Interaction (HCI) methods were employed in this research. These methods complemented each other in order to support our investigation and involved questionnaires, followed by participatory design sessions with user and iterative prototype design. The study was geographically limited to the United Kingdom (UK) and included only participants residing in the UK who are above 18 years of age and actively using smart home devices.

4.1 Questionnaire

The research began with an online questionnaire targeting smart home device users in the UK. This approach aimed to gather a broad range of insights from a diverse audience to identify common trends related to technology repair. Convenience sampling was used to recruit 50 participants from personal networks, including family, friends, university staff, and colleagues.

The questionnaire (Appendix A.3) consisted of a combination of closed-ended, Likert scale, and open-ended questions, focusing on key themes such as users' current repair and device maintenance practices, common issues with smart home devices, challenges and motivations related to repair, and preferred points of contact for seeking assistance. A pilot study with three users was conducted to test the clarity and validity of the survey, ensuring any ambiguities were addressed before distribution to the wider audience pool.

The results revealed a diverse demographic with 60% female respondents and the predominant age range of 29 to 48 years (58%). Most participants held higher education qualifications, with 36% possessing a bachelor's degree. Nearly half of the respondents (48%) owned more than five smart home devices, with many of these devices such as smartphones (90%), smart TVs (63%), and smart speakers (63%) in use for over three years. 41% of users identified as beginners in terms of smart home technology knowledge. Repair challenges were widespread, with difficulty finding repair information (49.01%), lack of technical skills (45.09%), and lack of manufacturer support (45.09%) as the top issues. Users highlighted unclear and outdated repair information, overly technical instructions, and a lack of detailed diagrams as major obstacles. To overcome these issues, many relied on YouTube videos as their primary resource for repair guidance, emphasising a preference for accessible, user-friendly platforms to supplement inadequate repair manuals. Cost-saving was the primary repair motivator (54%), while environmental concerns played a lesser role. Most favoured repair-friendly designs (82%) and better manufacturer support (67%). Here, maintenance relied heavily on automated updates (69%) and apps (45%), with a demand for more advanced lifecycle management tools. Overall, the findings emphasise users' willingness to repair their smart devices, tempered by barriers such as inaccessible repair information and technical complexities. This highlights the need for improved resources and stronger manufacturer support.

4.2 Participatory Design Session

A participatory design session was conducted to build on insights from the initial questionnaire by engaging participants in hands-on activities that explored their needs and preferences for potential solutions. This methodology encouraged collaboration by involving users as co-designers. Six participants were recruited from 18 individuals who expressed interest after the initial survey. The 2-hour online session, conducted via Microsoft Teams with Miro for collaboration, consisted of three sessions with two participants each. The workshop featured three hands-on activities: first, participants listed five smart home devices they owned and described their repair decisions for each (Figure 9); second, they brainstormed ideas to encourage self-repair of smart devices, grouped these ideas based on common themes, and prioritised them by urgency and

importance (Figure 10); finally, they translated these ideas into a mobile application mock-up, developing tangible app features (Figure 11, Figure 12). The session began with an icebreaker, followed by discussions about participants' experience with smart home devices and repair, which led into the ideation phase for designing potential solutions. With the participant's consent (regulated through the University's Ethics processes), the sessions were recorded, and notes were taken to capture any key insights that emerged during the discussions. Materials prepared for the session included medium-fidelity wireframes and interview questions. Post-session, data analysis involved thematic analysis and affinity diagramming to identify key themes related to participants' experiences and preferences. This approach provided rich insights while remaining accessible and efficient, with themes visualised by grouping related ideas and prioritising them based on relevance.

The themes were triangulated with the survey findings, resulting in that Table 1 presents the themes evolved from data analysis, supporting quotations, corresponding questionnaire results and overall findings.

4.3 Key Barriers to Repair

The study highlighted three major barriers to repair: users struggled to access clear and reliable repair information, as manufacturer guides were often vague, overly technical, or outdated, leading many to rely on YouTube tutorials. Additionally, a lack of technical skills and insufficient manufacturer support made repairs more challenging, emphasising the need for step-by-step guides and community-driven assistance. High repair costs and limited access to spare parts further discouraged repair efforts, as users faced difficulties in finding affordable components and reliable services.

These insights were incorporated into the mobile application's design to develop a smart care solution aimed at enhancing repairability and lifecycle management for smart home devices. The mobile application aims to support the circularity of smart home devices by providing users with tools to troubleshoot, repair, and maintain their devices, extending their useful life and reducing electronic waste. It encourages a more sustainable approach by facilitating repair, reuse, and the sharing of resources, thus contributing to the broader goals of sustainability and circular economy practices. However, the solution currently lacks a systemic view of repair that involves multiple stakeholders such as manufacturers, technicians, third party repair services etc. which will be addressed in future work, recognising the limitations imposed by planned obsolescence in current device designs. The app does not claim to single-handedly resolve systemic repair challenges. Instead, it serves as a facilitator within a broader repair ecosystem, complementing policy efforts like Right to Repair. Systemic barriers, such as manufacturers' reluctance to provide comprehensive repair information, proprietary designs that restrict access to repair tools or parts, and a lack of standardised repair procedures, are significant challenges that hinder the widespread adoption of repair practices. Ultimately, the solution envisions a future where devices are manufactured to be easily repaired and reused, promoting long-term sustainability.

4.4 Future Work: Prototype Design

The prototype design began with an iterative design approach to ensure the prototype closely aligned with users' needs. As shown in Figure 1, the design evolved through three iterations, each refined based on user feedback. The first iteration was developed using survey insights Table 1, where users highlighted the need for a centralised view of smart devices. Most users (68.62%) preferred automatic updates and alerts, while 29.41% opted for manual checks. The priorities for device health monitoring were automated alerts (47%), real-time performance monitoring (24%), and historical data (18%) (Table 1). However, participatory design sessions using this iteration of the prototype revealed that the comprehensive dashboard presented too much information, making it difficult for users to interpret the health score or act on it effectively. To address these concerns, the second iteration introduced a minimalist design, reducing cognitive load by presenting only essential information, such as device status and location, in a clear and accessible layout. Users appreciated the simplicity but suggested emphasising critical information requiring immediate attention. In the third iteration, a color-coded interface was implemented to prioritise critical alerts, ensuring urgent issues were easily identifiable.

The user journey starts with a centralised dashboard (Figure 1-Iteration 3) that provides an overview of all devices, displaying fault indicators and alerts to help users identify and address issues. Clicking an alert reveals diagnosed faults, their locations, and a detailed description, including severity, repair time, complexity, benefits, and step-by-step self-repair instructions. Users also receive tool procurement details and a quick-access button to connect with a technician. Figure 8 illustrates this fault diagnosis and resolution process. The fault report includes issue descriptions, severity levels, and actionable insights. Users are supported through video and text repair guides, tool and spare part recommendations, and a 'Chat with Technician' feature for live troubleshooting (Table 1). Additional details like model number, warranty status, and software version can be shared with a technician effortlessly. Users can also connect with community members for troubleshooting, saving time and costs. The AI-driven predictive maintenance feature (Figure 5, Figure 6) alerts users to upcoming maintenance needs and forecasts potential faults, ensuring proactive device management.

Future work will focus on conducting additional participatory design sessions to refine the application and develop another iteration based on user feedback. The prototype design will then undergo further evaluation.

5 Conclusion

This paper underscores the critical need to enhance the repairability and lifecycle management of smart home devices as a means to mitigate e-waste. The literature review highlights broader concerns surrounding e-waste and positions technology repair and maintenance as potential solutions, drawing upon circular economy principles. The paper further contextualises this emergent problem within the Human Computer Interaction (HCI) field, identifying both previous work and existing gaps. It then illustrates the genesis of design of a mobile application that empowers users in the repair and lifecycle management of smart home devices, addressing a significant gap in sustainable consumption practices.

Table 1: Research Findings

Themes	User Quotes	Questionnaire Results	Findings
First point of contact for seeking repair help	“My dad’s an electrician, so if I get stuck on these things, I don’t know what I’m doing, I will start sending my device photos and being like. What do you make of this?”	35.29% turned to online forums, 23.52% consulted YouTube, and 13.72% sought help from tech-savvy family or friends.	The findings reveal that integrating community support forum within the app could provide users with solutions to common repair problems. (Figure 8)
Challenges faced with repair process	“It generally takes me a lot of googling to figure out how to fix it. It is time-consuming to find the correct manual or instruction videos that work and help.”	Difficulty finding repair information (49.01%), lack of technical skills (45.09%), insufficient manufacturer support (45.09%), high repair costs (37.25%), and limited spare parts (29.41%).	Structured, step-by-step guided repair tutorials (video/text format) with simplified visuals and accessible spart parts links could enhance repair experience. (Figure 8)
Challenges with repair guides	“The manufacturers’ repair guides are completely and utterly useless. They don’t go beyond ‘have you switched it off and on again’?”	Lack of detailed diagrams or images (37.5%), overly technical information (31.25%), unclear instructions (31.25%), outdated manuals (31.25%), and 25% struggled to find relevant manuals.	Interactive repair guides with clear and simple schematics, augmented reality and digital twin technologies could improve user engagement and repair success rates.
Preference for repair over replacement	“If the phone had come with a repair guide or even an online PDF covering common repairs, it would have made things a lot easier since those were the challenges I faced.”	82.35% preferred repair if devices had simple designs, 66.67% supported by manufacturers, and 58.82% had access to affordable spare parts.	Manufacturer verified repair tools, spare parts information within the app could encourage more users to repair rather than replace their devices. (Figure 8)
Useful feature recommendation	“I really want the identify the problem thing because that will make my personal life easier.”	Automated alerts and notifications (47%), real-time performance monitoring (24%), and historical performance data (18%).	Integrating device performance monitoring, fault reminders, and historical logs could enhance user awareness and proactive maintenance. (Figure 2, Figure 4)
Visual media	“There’s always someone who’s done a similar thing and put it up on YouTube. That tends to be more helpful than anything from the manufacturers.”	YouTube tutorials were found more useful than manufacturer-provided materials, serving as a key resource for self-repairs.	Integrating curated video tutorials tailored to specific devices and repairs within the app could enhance accessibility and ease of learning. (Figure 8)
Product details	“I have to find my 5 to 6-year-old email to retrieve product details from my initial purchase.”	Users struggle to fetch product information such as model numbers, warranty details, and error codes when needed.	Centralising product information storage within the app could provide users with quick and easy access to essential device details. (Figure 3, Figure 7)
Contacting manufacturer support	“Long phone queues and unhelpful FAQs eventually lead to needing to call the helpline, which is frustrating and time-consuming.”	Users were dissatisfied with service centre support, citing long wait times, delayed responses, and frustration with phone queues.	Reducing reliance on phone queues by integrating live chat or direct messaging with technicians could significantly improve user satisfaction. (Figure 8)
Difficulty sourcing parts and information	“I don’t even know where you’d start to contact a technician or whether replacement parts for an Alexa can even be ordered.”	Participants reported significant challenges finding correct parts and repair instructions for their devices.	Providing technician assistance, repair tools, and spare parts information through the app could address users’ difficulties in sourcing components. (Figure 8)
Inaccessibility	“More accessibility support for people with disabilities and people who cannot speak the main service language.”	Repair processes were found inaccessible for less tech-savvy individuals, especially older users, due to complex instructions and technical knowledge barriers.	Implementing voice-assistant-enabled hands-free interaction and simplified language options could make the repair process more inclusive.

^a This includes findings obtained from both survey and participatory design sessions.

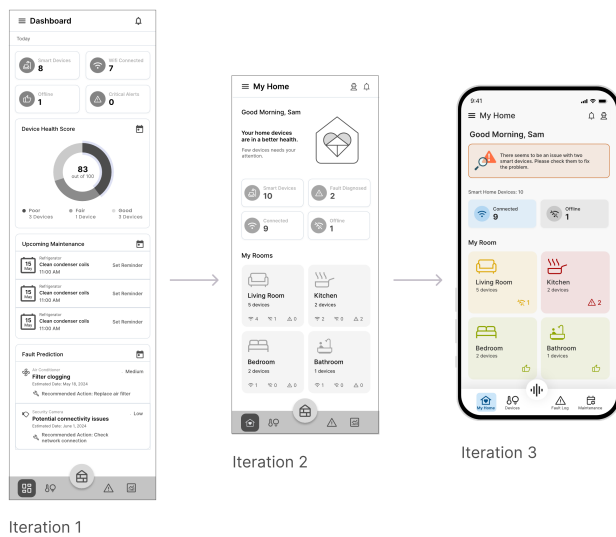


Figure 1: Home Screen Iterations

Utilising qualitative, formative research and prototyping methods, the design presented here provides practical and user-friendly solutions for the maintenance of everyday domestic smart devices. This work warrants further design refinement through user participation followed by evaluation of the design decisions employed through user feedback. By incorporating user insights into the design of a mobile app, this study aims to offer actionable solutions that promote sustainable consumption. Unlike existing platforms such as iFixit, which primarily offer static repair guides, this app integrates real-time diagnostics, AI-driven predictive maintenance, and community-driven troubleshooting. By facilitating access to structured repair information, step-by-step guides, and technician support, the solution aims to empower users to maintain their devices more effectively while addressing key repair barriers identified in this research. While this solution does not resolve systemic barriers alone, it provides a practical tool that empowers users, complements Right to Repair efforts, and contributes to a circular economy framework. Ultimately, this research seeks to foster sustainable practices and serve as a model for integrating reparability into consumer technology ecosystems.

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A APPENDICES

A.1 Mobile application prototype screens

A.1.1 Home Screen

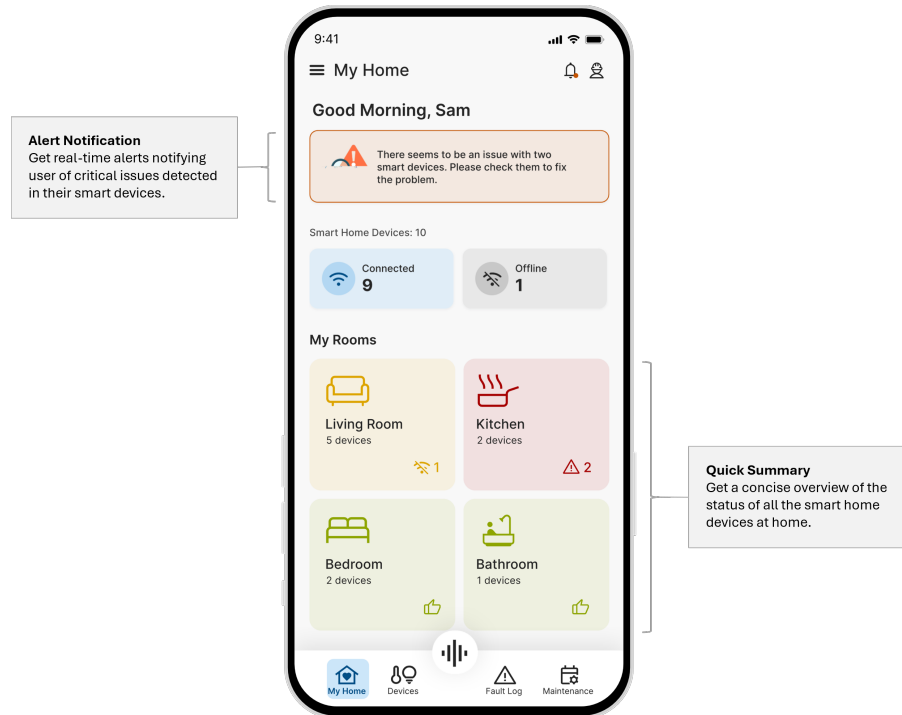


Figure 2: Home Screen

A.1.2 Smart Devices (All rooms and kitchen) Screens

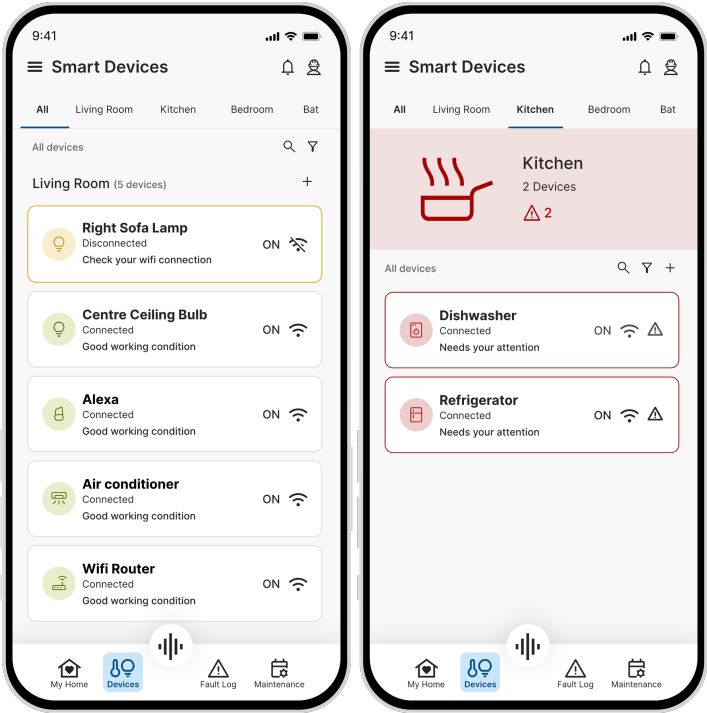


Figure 3: List of Smart Devices

A.1.3 Fault Log (History and Prediction) Screens

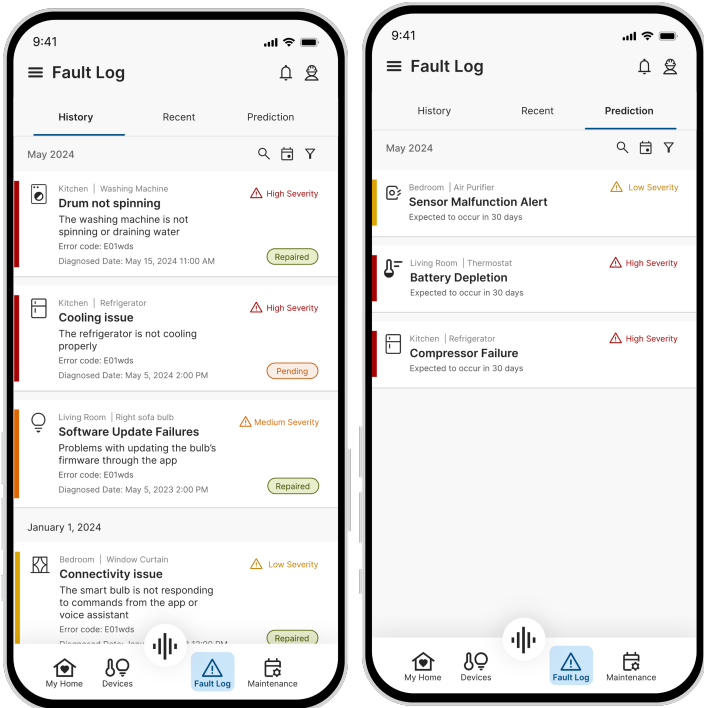


Figure 4: Fault Log Screens

A.1.4 Fault Prediction

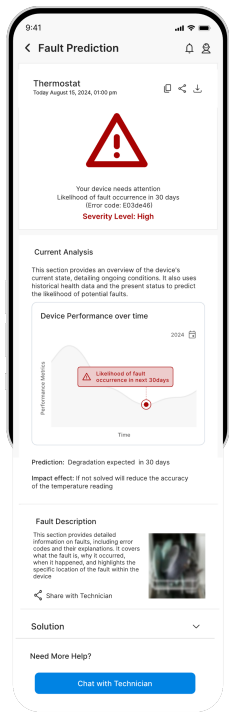


Figure 5: Fault Prediction

A.1.5 Maintenance

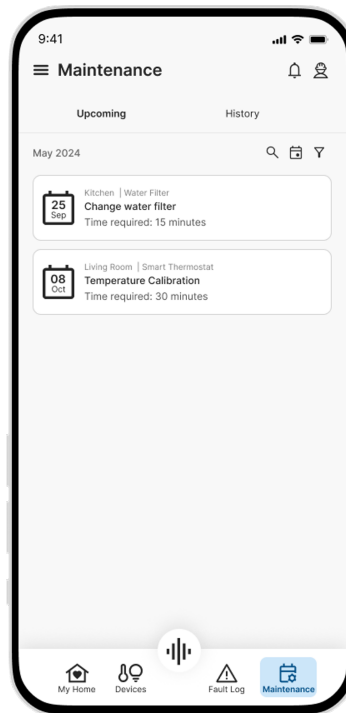


Figure 6: Upcoming Maintenance

A.1.6 Device Information

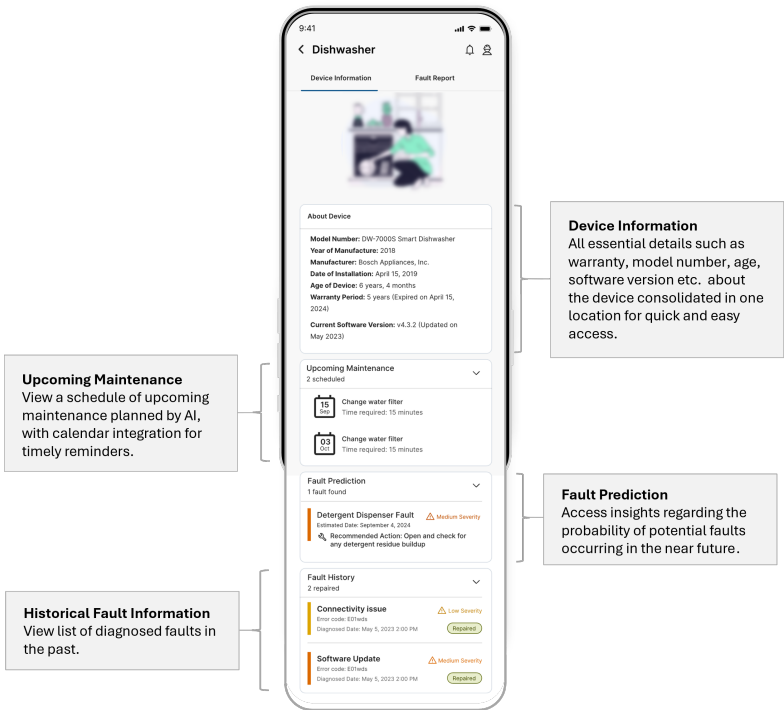


Figure 7: Smart Device Information

A.1.7 Fault Report

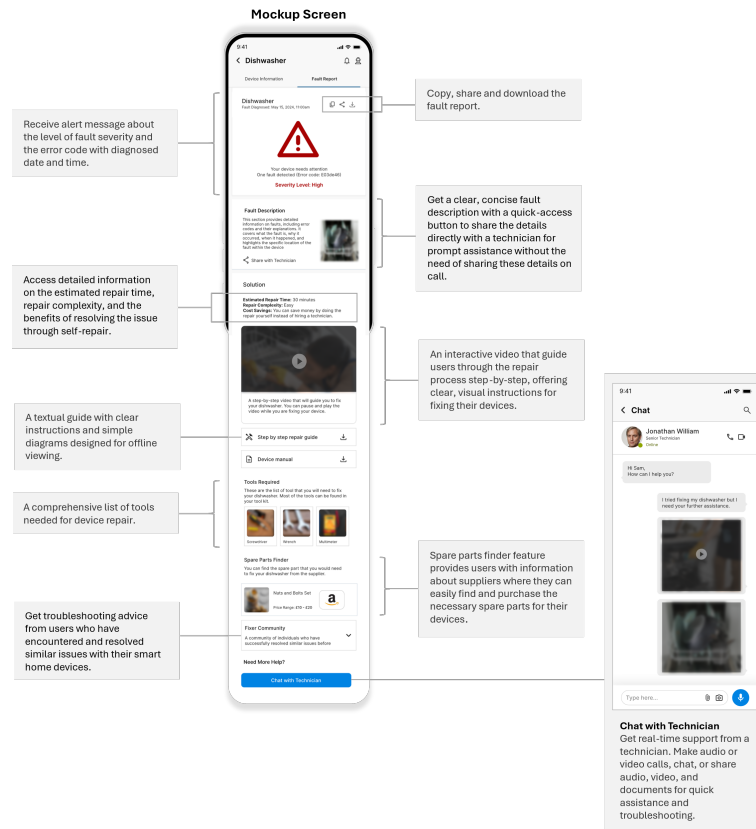


Figure 8: Fault Report

A.2 Screenshots of participatory design sessions

A.2.1 Group interview activity

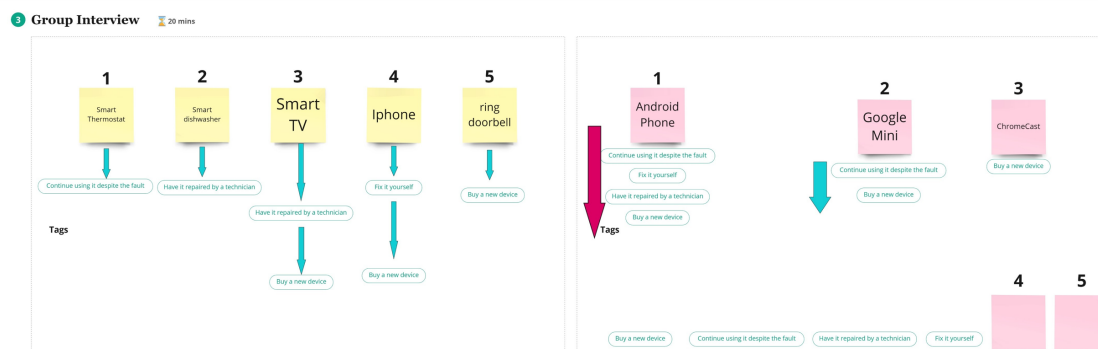
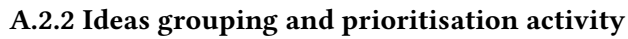


Figure 9: Group Interview Activity



A.2.4 Screen design ideation activity – participant 2

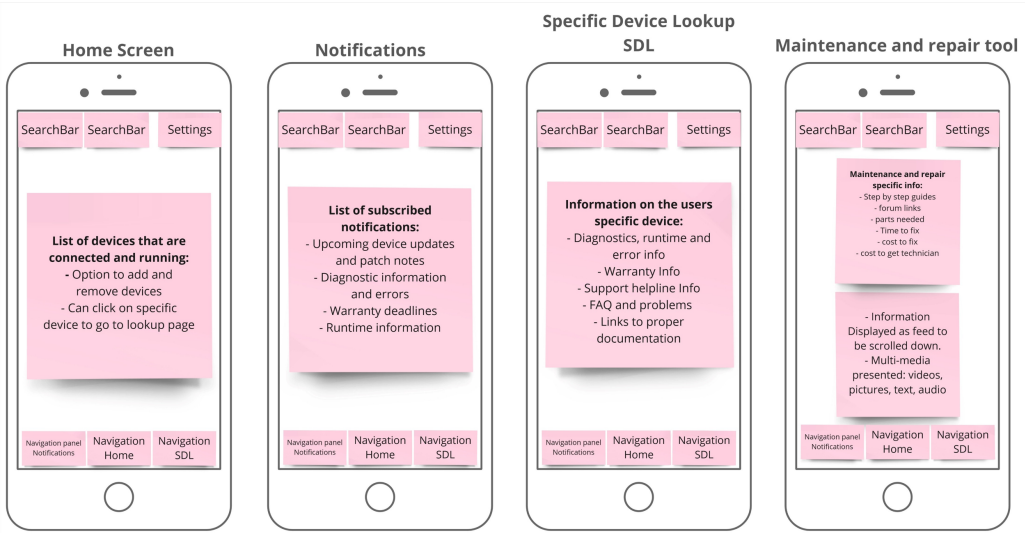


Figure 12: Screen Design – Participant 2

A.3 Questionnaire for smart home device users

Part 1: Demographic Information

2. Age *

☐ 18 to 28

☐ 29 to 38

☐ 39 to 48

☐ 49 to 58

☐ 59 years and above

3. Gender *

☐ Male

☐ Female

☐ Prefer not to say

4. Education Level *

☐ GCSE or equivalent

☐ A-levels or equivalent

☐ College

☐ Higher National Diploma or equivalent

☐ Bachelor's degree

☐ Master's degree

☐ PhD or equivalent

5. Household size *

☐ 1

☐ 2 family members

☐ 3 to 4 family members

☐ more than 5 family members

Part 2: Smart Device Information

6. How many smart devices do you have at home? *

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ More than 5

7. How frequently do you interact with your smart devices? *

☐ Multiple times a day

☐ Once a day

☐ A few times a week

☐ Once a week

☐ Rarely (Once a month/year)

8. Since how long have you been using the smart devices at your home? *

☐ Less than 6 months

☐ 6 months to 1 year

☐ 1 to 2 years

☐ 2 to 3 years

☐ More than 3 years

9. Which of the following smart devices do you currently use at home? *

☐ Smart Speakers (Google Home, Amazon Echo & Alexa, Apple's HomePod etc)

☐ Smart Plugs

☐ Smart Bulbs

☐ Smart Door Locks

☐ Smart Thermostats

☐ Smart TVs

☐ Smart Appliances such as washing machine, Fridge etc

☐ Smart Vacuum cleaner

☐ Smartphones

☐ Smartwatches (Fitbit, Apple watch etc)

☐ Smart shoes

☐ All of the above

☐ Other

10. How would you rate technology knowledge and expertise using smart devices? *

☐ Beginner (someone who has just started learning, knows basic troubleshooting techniques, familiar with common issues and solutions but typically rely on simple guides or instructions for repairs)

☐ Intermediate (Someone who has a good grasp, can handle a wider range of issues, perform more complex repairs and are comfortable with some level of disassembly and part replacement)

☐ Advanced (someone who has extensive experience, can tackle most types of repairs, skilled at diagnosing and solving intricate problems)

☐ Expert (someone who is highly skilled and experienced, possesses comprehensive knowledge of device systems and can handle nearly any issue)

☐ Novice (someone who is very new to home IoT device repair, unfamiliar with basic troubleshooting and repair practices and usually need considerable guidance)

Part 3: Smart Device Repair

11. How do you find out that your smart home devices needs repair? *

☐ Error messages/alerts from the device

☐ Physical damage

☐ Device performance issues

☐ System updates notification

☐ Other

12. What common issues do you encounter with your smart devices? *

☐ Connectivity problems (e.g., Wi-Fi or Bluetooth issues)

☐ Device not responding

☐ Performance slowdowns

☐ Device sync issue

☐ Internal hardware malfunctioning

☐ Battery or power issues

☐ Difficulty with setup or configuration

☐ Incompatibility with other devices or systems

☐ Poor customer support or lack of documentation

☐ Frequent updates or software incompatibility

☐ Physical damage or wear and tear

☐ Security concerns (e.g., data breaches or vulnerabilities)

☐ Other

Figure 13: Questionnaire Part 1: Demographics, Smart Device Usage and Challenges

13. Who is your first point of contact when seeking help for repair? *

☐ Manufacturer support

☐ Professional repair services such as Repair Cafés

☐ Tech-savvy family or friends

☐ Online forums/communities

☐ YouTube

☐ Mgeni

☐ Other

14. How often do you try to repair your smart home devices when they not functioning properly? *

☐ Always

☐ Sometimes

☐ Rarely

☐ Never

15. What resources or tools do you use for repair? *

☐ Online repair guides

☐ Manufacturer support

☐ Professional repair services

☐ DIY repair kits

☐ Community forums

☐ YouTube videos

☐ All of the above

☐ Other

16. How repair guides enabled you to do repairs you wouldn't have been able to do otherwise? *

☐ Yes

☐ No

17. How helpful do you think are the repair guides provided with the smart home devices? Rate on a scale of 1 to 5, where 1 means "Not at all helpful" 2 means "Somewhat helpful" 3 means "Moderately helpful" 4 means "Very helpful" 5 means "Extremely helpful" *

1

2

3

4

5

18. What issues do you encounter with the repair guides? *

☐ Instructions are unclear or confusing

☐ Lack of detailed diagrams or images

☐ Information is too technical

☐ Manuals are difficult to find

☐ Instructions are outdated

☐ Manuals do not cover common problems

☐ Steps are too complicated to follow

☐ Other

19. How you purchased insurance coverage for your smart device? *

☐ Yes

☐ No

20. Which of the following factors motivate you most to repair your smart home devices? *

☐ Cost savings

☐ Environmental concerns

☐ Personal satisfaction

☐ Preserve sentimental value

☐ Lack of affordable replacements

☐ Device with an active warranty

☐ Device insurance cover

☐ Other

21. How motivated are you to repair your smart home devices for the following reasons? Please rate each item on a scale from 1 to 5, where 1 means "Not at all motivated" 2 means "Slightly motivated" 3 means "Moderately motivated" 4 means "Very motivated" 5 means "Extremely motivated" *

1

2

3

4

5

Cost savings

☐

☐

☐

☐

☐

Environmental concerns

☐

☐

☐

☐

☐

Personal satisfaction

☐

☐

☐

☐

☐

Device sentimental value

☐

☐

☐

☐

☐

Lack of affordable replacements

☐

☐

☐

☐

☐

Device with an active warranty

☐

☐

☐

☐

☐

Device insurance cover

☐

☐

☐

☐

☐

22. How would you rate your past experience of repairing a smart home device at the service centre? *

☐ ☆

☐ ☆

☐ ☆

☐ ☆

23. What are the reasons behind your rating given in the previous question? *

24. What challenges do you face in this journey of getting your smart home device repaired? *

☐ Lack of technical skills

☐ High cost of repairs

☐ Difficulty finding repair information

☐ Availability of spare parts

☐ Lack of support from the manufacturer

☐ Other

25. At what point do you decide to replace your smart device? *

☐ When a newer model is released

☐ When the current device no longer meets my performance expectations

☐ When the device requires frequent repairs

☐ When the device experiences a major malfunction or breakdown

☐ When the device is beyond economical repair

☐ When the manufacturer no longer supports the device with updates or parts

☐ When I want to upgrade to a device with new features

☐ When the device warranty expires

☐ Other

26. What features would make you more likely to repair your smart home devices rather than replace them? *

☐ Availability of repair guides

☐ Access to affordable spare parts

☐ Manufacturer support for repairs

☐ Simple design for easy repair

☐ Community support/forums

☐ Other

27. Do you believe that extending the lifecycle (using devices for longer time) of smart devices can have a positive impact on the environment? Please rate your agreement using the scale below. *

☐ ☆

☐ ☆

☐ ☆

☐ ☆

Figure 14: Questionnaire Part 2: User Behaviors, Barriers, and Motivations in Smart Device Repair

Part 4: Product Maintenance

28. How often do you check for updates on your smart home devices? *

☐ Always

☐ Sometimes

☐ Rarely

☐ Never

29. How do you typically check the health status of your smart devices? *

☐ Manufacturer's app

☐ Third-party monitoring app

☐ Manual checks

☐ Other

30. How do you manage to take care of all smart devices at home? *

☐ Individually check each device manually

☐ Use a centralized system or hub to monitor and manage devices

☐ Schedule regular maintenance tasks using a digital calendar or planner

☐ Rely on automatic updates and alerts from the devices themselves

☐ Utilize a smart home management app for oversight and control

☐ Employ professional maintenance services for comprehensive care

☐ Other

31. What challenges do you face while maintaining your smart home devices? *

☐ Keeping track of updates

☐ Identifying performance issues

☐ Lack of time

☐ Technical difficulties

☐ Coordinating updates and maintenance

☐ Lack of integration between devices

☐ High cost of management tools

☐ Other

Part 5: Lifecycle Management of Smart Devices at Home

Lifecycle management involves:
1. Setup: Initial setup and configuring the device.
2. Maintenance: Regular checking performance, health, and updates.
3. Management: Managing settings, monitoring usage, and optimizing performance.
4. Tracking: Recording usage history and maintenance to guide repairs and replacement.
5. End-of-life: Deciding when to replace, dispose of, or recycle the device, and addressing malfunctions through repairs or upgrades.

32. Do you use any lifecycle management tools or platforms to track the performance and health of your smart devices? *

☐ Yes

☐ No

33. How would you rate the effectiveness of the lifecycle management tools you use? *

☐ Very effective

☐ Somewhat effective

☐ Not effective

34. What features do you find most useful in lifecycle management tools for smart devices? *

☐ Real-time performance monitoring

☐ Automated alerts and notifications

☐ Historical performance data

☐ Predictive maintenance insights

☐ Other

35. Would you find lifecycle management tools more effective if they included the following improvements? *

☐ Enhanced user interface

☐ More automation features

☐ Better integration with other tools

☐ Improved reporting and analytics

☐ More customizable options

☐ Other

36. How likely are you to use a centralized platform that offers comprehensive lifecycle management for all your smart home devices? *

☐ ☆

☐ ☆

☐ ☆

☐ ☆

37. Do you have any additional comments or suggestions for improving the repairability and lifecycle management of smart home devices? *

Figure 15: Questionnaire Part 3: User Practices and Challenges in Smart Device Maintenance